Skills Summary

Derivatives of Inverse and Inverse-Trigonometric Functions

Use this reference while completing your worksheet or when studying.

Every rule below comes from one idea: if y is the inverse of something you can already differentiate, implicit differentiation turns the known derivative upside down. That is why each arc-formula is a reciprocal, and why $(f^{-1})'$ is a reciprocal too.

1. Arc-functions with the chain rule

With inner function $u = u(x)$:

$$\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}} \qquad \frac{d}{dx}\arccos u = \frac{-u'}{\sqrt{1-u^2}} \qquad \frac{d}{dx}\arctan u = \frac{u'}{1+u^2}$$

The arccos formula is the arcsin formula with a minus sign; the arctan denominator has *no* radical. Substitute u into the denominator *before* multiplying by u'.

Example: Differentiate $y = \arcsin(4x)$

Step 1. This is an arc-function of an inner function, so use the arcsin formula with $u = 4x$ and carry the chain-rule factor u'.

Step 2. $u' = 4$ and $u^2 = 16x^2$, so $y' = \dfrac{4}{\sqrt{1-16x^2}}$. $\Rightarrow$ $\boldsymbol{y' = \dfrac{4}{\sqrt{1-16x^2}}}$

Try it: Differentiate $y = \arctan(5x)$

check: $\boldsymbol{\dfrac{5}{1+25x^2}}$

2. The inverse-function derivative

Rule: if f is one-to-one and $f(b) = a$, then $\left(f^{-1}\right)'(a) = \dfrac{1}{f'(b)}$.

Order of work: (1) find the input b that f sends to a — usually by inspection or a quick solve, (2) differentiate f, (3) evaluate f' **at** b, (4) take the reciprocal. Evaluating f' at a instead of b is the classic lost mark.

Example: $f(x) = x^3 + x + 2$. Find $\left(f^{-1}\right)'(4)$.

Step 1. The question asks for the inverse's slope at an output value, so find the matching input first and then reciprocate f' there.

Step 2. $f(1) = 4$, so $b = 1$; $f'(x) = 3x^2 + 1$, so $f'(1) = 4$. $\Rightarrow$ $\left(f^{-1}\right)'(4) = \boldsymbol{\frac{1}{4}}$

Try it: $f(x) = x^3 + 2x$, and $f(1) = 3$. Find $\left(f^{-1}\right)'(3)$.

check: $\boldsymbol{\frac{1}{5}}$

3. Products with an arc-function

Product rule: $(fg)' = f'g + fg'$, with the arc-derivative used for whichever factor is the arc-function.

Differentiate the arc-factor *in place* — do not try to simplify $\arctan x$ away first. The answer usually keeps an arc-function term *and* a rational term; that is normal, not unfinished.

Example: Differentiate $y = x^2 \arctan(x)$

Step 1. Two functions are multiplied and neither is a constant, so this is the product rule with the arctan derivative supplying g'.

Step 2. $f = x^2$, $f' = 2x$; $g = \arctan x$, $g' = \dfrac{1}{1+x^2}$. $\Rightarrow$ $\quad y' = \boldsymbol{2x\arctan(x) + \dfrac{x^2}{1+x^2}}$

Try it: Differentiate $y = x \arcsin(x)$

check: $\boldsymbol{\arcsin(x) + \dfrac{x}{\sqrt{1-x^2}}}$

4. Second derivatives of arc-functions

Method: write the first derivative as a **power**, then use the chain rule again:

$$\frac{1}{1+x^2} = (1+x^2)^{-1} \qquad \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-1/2}$$

Negative and fractional exponents make the second differentiation ordinary power-and-chain work — no quotient rule needed.

Example: Find y'' for $y = \arcsin(x)$

Step 1. The first derivative is a radical in the denominator, so rewrite it as a power before differentiating a second time.

Step 2. $y' = (1-x^2)^{-1/2}$, so $y'' = -\frac{1}{2}(1-x^2)^{-3/2}(-2x) = x(1-x^2)^{-3/2}$. $\Rightarrow$ $\quad y'' = \boldsymbol{\dfrac{x}{(1-x^2)\sqrt{1-x^2}}}$

Try it: Find y'' for $y = \arccos(x)$

check: $\boldsymbol{\dfrac{-x}{(1-x^2)\sqrt{1-x^2}}}$